

LEARNING MODULE

5

Company Analysis: Past and Present

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe the elements that should be covered in a thorough company research report
☐	determine a company's business model
☐	evaluate a company's revenue and revenue drivers, including pricing power
☐	evaluate a company's operating profitability and working capital using key measures
☐	evaluate a company's capital investments and capital structure

INTRODUCTION

1

An insightful and well-written company research report helps investors understand a company and make better investment decisions about the company's securities. This module is the first of three that will provide a framework to prepare a company research report by applying methods covered in previous modules to assess a company's business model, financial performance, and financial position.

LEARNING MODULE OVERVIEW



- Company research reports analyze a company's past and present financials, its industry and competitors, and forecast its future financial statements. Reports end with a valuation, an investment recommendation, and investment risks.
- The first step of company analysis requires an understanding of the issuer's business model, for which analysts rely on both issuer and third-party information sources.
- An understanding of the issuer's business model and analysis of historical financial statements will allow the analyst to identify key drivers of revenues, profitability, cash flows, and financial position.

- Revenue analysis can be done using a bottom-up or top-down approach. A bottom-up approach breaks down revenues into drivers such as sales volumes and prices, by product line or segment. A top-down approach expresses revenue as a function of drivers such as market share, market size, and GDP growth.
- While a company can change its prices at will, its ability to do so successfully (i.e., not causing a loss of volume) and relative to costs is driven by the company's pricing power. Pricing power is primarily a function of industry structure and competitive strategy.
- Cost analysis assesses a company's profitability and working capital management. Analysts calculate and interpret several cost and profitability measures, including gross, operating, and net margins.
- While a fixed versus variable cost analysis is useful, the analysis is often limited by issuer disclosures and accounting standards that emphasize cost reporting by function or nature.
- The degree of operating leverage measures the sensitivity of operating profit to changes in sales. Operating leverage is primarily driven by the fixed and variable cost composition of the issuer's operating expenses.
- The degree of financial leverage measures the sensitivity of net income to changes in operating income. Financial leverage is primarily driven by the issuer's capital structure.

LEARNING MODULE SELF-ASSESSMENT



These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. Identify two elements that are common to *all* company research reports and two elements that are common to *initial* research reports.

Solution:

Common to all research reports:

- Analyst's investment recommendation and target buy or sell prices
- Risks such as evaluation of material downside and upside risks

Common to initial research reports:

- Discussion of issuer's business model and strategy, and explanatory charts and figures that disaggregate revenues and profits by product or geography
- Detailed financial analysis and models
- Industry overview and competitive positioning such as industry size, growth rate, market share trends, and industry profitability. Evaluation of the company's competitive position and strategy in each product line or segment is also common.

2. Identify three information sources that analysts commonly use to determine and analyze a company's business model.

Solution:

- Regulatory filings if the issuer is public, especially the annual and quarterly reports
- Investor events and presentations by the issuer
- Visiting the company's properties and/or websites

3. Last year, a distributor of dental care products earned EUR800 million in revenues. Management estimated that its market share was 10%. Based on management's estimate, the market size for distribution of dental care products in this geography last year was *closest* to (in millions of EUR):

- A. 80
- B. 7,200
- C. 8,000

Solution:

C is correct.

$$\text{Market share} = \frac{\text{Revenue}}{\text{Market Size}}$$

Which can be rearranged to:

$$\text{Market size} = \frac{\text{Revenue}}{\text{Market Share}}$$

$$\text{Market size} = \frac{\text{EUR 800 million}}{10\%}$$

$$\text{Market size} = \text{EUR 8,000 million (8 billion)}$$

4. A beverage manufacturer recently introduced a new line of healthy tea-based drinks. The average selling price was USD20 a case last year and the company sold 2 million cases. This year, an analyst forecasts sales of 3 million cases and an increase in the average selling price—from a reduction in discounts and promotions—of 5%. The analyst's forecast for net sales growth this year over the prior year is *closest* to:

- A. 50%
- B. 55%
- C. 58%

Solution:

C is correct.

$$\text{Net sales} = \text{average selling price} \times \text{cases sold}$$

$$\text{Net sales last year} = \text{USD20} \times 2 \text{ million} = \text{USD40 million}$$

$$\text{Net sales forecast this year} = (\text{USD20} \times 1.05) \times 3 \text{ million} = \text{USD63 million}$$

$$\text{Forecasted net sales growth rate} = 63/40 - 1 = 0.575 \text{ or } 58\%$$

5. Impression Ltd. is a fictional company that designs, manufactures, and sells skin care and beauty products. Identify whether each expense line below

from Impression Ltd.'s last annual income statement is *most likely* a fixed or variable cost.

I. Amortization of acquired intangible assets	Fixed	Variable
II. Interest expense	Fixed	Variable
III. Costs of goods sold	Fixed	Variable
IV. General and administrative expenses	Fixed	Variable
V. Sales commissions	Fixed	Variable

Solution:

I. Fixed. Amortization expense generally does not change with levels of sales volume, as definite-lived intangible assets are amortized on a straight-line basis over the assets' useful lives.

II. Fixed. Interest expense is a function of the quantity of debt and its interest rate, not sales volume.

III. Variable. Costs of goods sold are incurred when sales are made.

IV. Fixed. General and administrative expenses are usually not related to sales volume each year.

V. Variable. Sales commissions, a form of performance-based compensation for salespeople, are a function of sales and are therefore variable.

6. Based on the data below, the degree of operating leverage for Company XYZ for the year ended 31 December 20X1 is *closest* to:

Company XYZ: Statement of Income for the Year Ended (in millions of EUR)

	31 December 20X1	31 December 20X0
Revenue	9,707	9,256
Costs of goods sold	4,850	4,637
Selling, general, and administrative expenses	993	1,090
Research and development expenses	1,700	1,554
Other operating expenses	491	448
Interest expense	309	325
Other (income) expense	24	(71)
Income before income taxes	1,340	1,273
Provision for income taxes	295	255
Net income	1,045	1,018

A. 0.51

B. 1.08

C. 1.96

C is correct.

$$20X1 \text{ Operating Income} = 9,707 - 4,850 - 993 - 1,700 - 491 = 1,673$$

$$20X0 \text{ Operating Income} = 9,256 - 4,637 - 1,090 - 1,554 - 448 = 1,527$$

$$DOL = \% \Delta \text{ Operating Income} / \% \Delta \text{ Sales}$$

$$DOL = (1,673/1,527 - 1) / (9,707/9,256 - 1)$$

$$DOL = 1.96$$

A is incorrect, as it inverts the equation; it is the ratio of the change in sales to the change in operating income.

B is incorrect; it is the ratio of the percentage change in pre-tax income, rather than operating income, to the percentage change in sales.

7. If a company reported trailing 12 months' operating cash flow of USD150 million, current assets excluding cash of USD40 million, and current liabilities of USD60 million, the company *most likely*:

- A. was inefficient in managing its financing.
- B. used trade credit to manage working capital.
- C. over-extended itself and is at risk of bankruptcy.

Solution:

B is correct. Negative net working capital typically indicates that the company uses suppliers' trade credit to lengthen days payable, while also efficiently managing inventory days and days sales outstanding.

A is incorrect. Negative working capital is a source rather than a use of financing and gives an issuer more financial flexibility, as capital is not tied up in working capital.

C is incorrect. There is not enough information here to determine bankruptcy risk, but positive operating cash flows and negative net working capital are not bankruptcy risk indicators.

8. A company's management may choose *not* to use financial leverage because the company has:

- A. an investment-grade credit rating.
- B. a high degree of operating leverage.
- C. a lower net debt to EBITDA ratio than its peers.

Solution:

B is correct. The degree of financial leverage and the degree of operating leverage together equal the degree of total leverage in the business. If the company already has a high degree of operating leverage, using financial leverage may increase total leverage and risk to too high a level.

A is incorrect. An investment-grade credit rating generally indicates that the company could borrow economically, which would be a reason to use financial leverage.

C is incorrect. A lower net debt to EBITDA ratio than its peers is an indicator of borrowing capacity, which would be a reason to use financial leverage.

COMPANY RESEARCH REPORTS

2

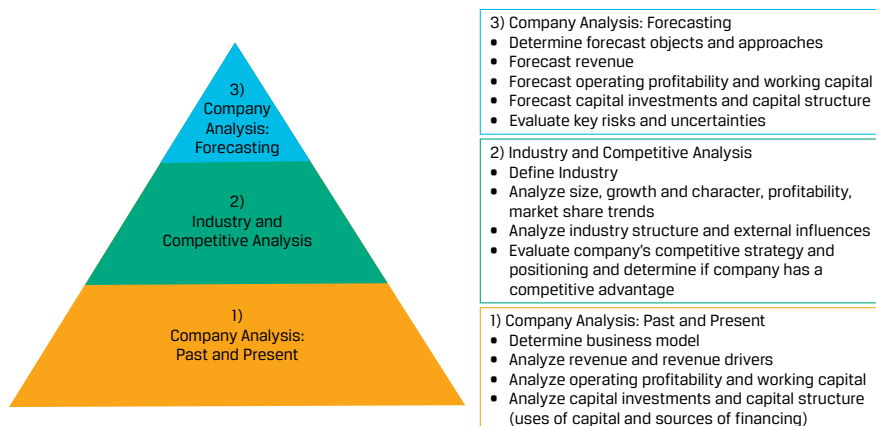


describe the elements that should be covered in a thorough company research report

Analysts value and make investment recommendations on issuers' equity securities using scenarios of future earnings, cash flows, and financial position. These future scenarios are structured in the form of financial statements and are known as financial statement models. While a financial statement model is quantitative, it is not a mathematics problem to solve with a correct answer, but rather a quantitative expression of an analyst's forward-looking views. These views should be based on supporting evidence and justified by analysis.

The process of forming and justifying a view of an issuer's future financial results and position involves company and industry analysis and is the subject of this module and the next two. Exhibit 1 introduces a framework for company and industry analysis. This module focuses on the foundational block Company Analysis: Past and Present.

Exhibit 1: Company and Industry Analysis Framework



Analysts present their company and industry analysis, as well as their valuation and investment recommendation, in a **company research report**. The structure, content, and tone of the report depend on the analyst's setting. Reports on a public issuer's equity securities for distribution to external clients (a "sell-side report"), for example, often consist of an extensive initial report (an "initiating coverage" report, or "initiation") when the analyst begins covering the security, followed by shorter reports on specific topics or that make updates to a recommendation after the analyst receives new information or conducts additional analyses, particularly after the issuer reports financial results. Analysts' reports solely for internal distribution to an audience already familiar with the issuer might be far shorter and delivered verbally or in a handful of presentation slides.

An example of the structure of an *initial* company research report for external distribution is shown in Exhibit 2. The primary audience is those who are not already knowledgeable about the issuer or security.

Exhibit 2: Initial Company Research Report Elements

Front Matter	▪ Issuer name and security identifiers (e.g., symbol, CUSIP) ▪ Analysts' recommendation (buy, hold, sell) and target buy or sell prices ▪ Disclosures, disclaimers, and other legally required information
Recommendation	▪ Analysts' recommendation ▪ Summary of key reasons supporting the recommendation
Company Description	▪ Discussion of issuer's business model and strategy ▪ Explanatory charts and figures—for example, disaggregation of revenues and profits by product, geography, etc.
Industry Overview & Competitive Positioning	▪ Industry size, growth rate and key drivers, market share trends, profitability—historical and outlook ▪ Competitive analysis such as Porter's Five Forces ▪ Analysis of external industry influences such as political, economic, social, technological, legal, and environmental ("PESTLE" analysis) ▪ Evaluation of company's position and strategy in the industry
Financial Analysis and Model	▪ Evaluation of key drivers of revenue, costs, profitability, cash flows, and the issuer's uses and sources of capital ▪ Forecasts of key drivers and supporting discussion ▪ Historical and forecasted financial statements
Valuation	▪ Estimates of company and security values with target price ▪ Typically includes relative and present value approaches ▪ Discussion of key inputs and scenario and sensitivity analyses
Environmental, Social, and Governance (ESG) Considerations	▪ Evaluation of ESG indicators and risks ▪ Ownership structure and management composition ▪ Executive compensation
Risks	▪ Evaluation of material downside and upside risks and discussion of how they are considered in the financial analysis and valuation

THE CFA INSTITUTE RESEARCH CHALLENGE

For examples of initial company research reports, we encourage you to visit the Past Research Challenge Champions section of cfainstitute.org, which has winning reports and presentations from the research competition that CFA Institute and CFA Societies around the world host each year. Teams of university students are assigned a public company to research by a local CFA Society and compete on the quality of their research and modeling, valuation, and communication skills.

Subsequent company research reports are often shorter than initial reports. Their primary audience is those who are already familiar with the issuer or security and require an update based on new information and analyses or a change in the analyst's recommendation. Such a report for a company that recently reported quarterly financial results may be structured in the form shown in Exhibit 3. The structure of a subsequent report depends on the analyst's setting and the nature of the report.

Exhibit 3: Subsequent Company Research Report Elements

Front Matter	▪ Analysts' names ▪ Issuer name ▪ Security and exchange identifiers (e.g., symbol, CUSIP) ▪ Analysts' recommendation: buy, hold, sell ▪ Current security price and analysts' target price ▪ Disclosures, disclaimers, and other legally required information
Recommendation	▪ Analysts' recommendation ▪ Summary of changes to the recommendation and support explanations
Analysis of New Information	▪ Comparison of quarterly actual results with expected results ▪ Analysts' interpretation of new information and changes to forecasts ▪ Historical and updated forecasted financial statements
Valuation	▪ Estimates of company and security values ▪ Discussion of changes from valuation in prior report
Risks	▪ Update of risk factors from prior report with discussion of changes

QUESTION SET



1. Which of the following statements about initiation reports is true?

- A. The structure, content, and tone of the report are independent of the analyst's setting.
- B. The report makes updates to the recommendation after the issuer reports financial results.
- C. The primary audience is those who are not already knowledgeable about the issuer or security.

Solution:

C is correct. The primary audience is those who are not already knowledgeable about the issuer or security.

A is incorrect, because the structure, content, and tone of the report are dependent on the analyst's setting. Examples include whether the report is meant for internal or external distribution and whether it is an in-depth initiation report or a subsequent report updating the reader after a change in recommendation or the release of the issuer's quarterly results.

B is incorrect, because the initial report is to help those who are not already knowledgeable to understand the issuer or security. A subsequent research report is used to update the audience after the release of financial results.

2. If you read a research report that includes both a Porter's Five Forces analysis and a PESTLE analysis, then *most likely*:

- A. the authoring analyst is initiating coverage.
- B. the report is updating a recommendation after an earnings release.

- C.** the report is written by a firm's analyst who has covered the company for many years.

Solution:

A is correct. A Porter's Five Forces analysis and a PESTLE analysis are common elements in initiations and are less common in subsequent research reports, as they tend to evolve more slowly than other report elements such as financial analyses and models.

B and C are incorrect; these elements are common in subsequent rather than initial research reports.

3. The evaluation of a company's competitive strategy takes place before the:

- A.** analysis of revenue and revenue drivers.
B. analysis of operating profitability and working capital.
C. forecasting of revenue, operating profitability, and working capital.

Solution:

C is correct. Forecasting takes place only after company analysis and industry and competitive analysis are complete. The evaluation of a company's competitive strategy is part of the industry and competitive analysis.

A is incorrect, because the analysis of revenue and revenue drivers is part of the first stage of analysis, where the analyst studies the company's past and present financial statements. As we shall see in Lesson 3, analysts start by analyzing the revenue line items first.

B is incorrect, because the analysis of operating profitability and working capital typically takes place before the industry and competitive analysis.

The evaluation of the company's competitive strategy is usually part of the industry and company analysis, which takes place after completion of the analysis of operating profitability and working capital.

DETERMINING THE BUSINESS MODEL

3



determine a company's business model

Determining the business model is the first step in our industry and company analysis framework because it summarizes important drivers of an issuer's financial results and position, focuses the analyst on what requires further investigation, and should begin setting the analyst's expectations for the issuer. A discussion of the business model is usually in the first part of a company research report (see the "Company Description" section in Exhibit 2).

Recall that a business model describes a company's operations and includes the following elements, which analysts investigate by answering several key questions discussed in prior learning modules.

Business Model Element	Key Questions for Analysts
The product(s) or service(s) the company sells	What is the firm selling?
Customers and key customer groups	To whom does the company sell?

Business Model Element	Key Questions for Analysts
Sales channels, including customer acquisition and product/service delivery mechanisms	How does the company reach potential and current customers and how does it deliver products?
How the product(s) or service(s) are priced and paid for	How much does the company charge and how are prices and payment terms structured?
Resource, supplier, and partner relationships need to operate effectively	What does the company buy and rely on?

There are many corollary questions for each of the key questions, such as determining whether customers are few and concentrated, whether suppliers and resources are specialized or hard to replace, and so on. The answers to these questions are company specific, but the key questions to ask are common across industries and companies. Also, recall that many companies have a conventional business model, such as retailer or natural resource producer, which simplifies business model identification in practice. Analysts will often focus their analysis on the differences in a company's business model, if any, from a conventional model or those of its competitors.

Information sources that analysts use to answer these questions include the following, grouped by origin: the issuer, public third party, proprietary third party, and proprietary primary research.

- Issuer sources (available freely if the company is public)
 - Regulatory filings, especially the annual and quarterly reports
 - Quarterly or semi-annual earnings conference calls
 - Presentations and events for investors
 - Press releases
 - Issuer management, investor relations, or other personnel
 - Company website or properties that the analyst may be able to visit as either a customer or an investor. It is often useful to experience an issuer's and its competitors' products firsthand, though it is not always possible (e.g., pharmaceuticals).
- Public third-party sources
 - Free industry white papers or analyst reports from a consultancy
 - Economic or industry indicators from governments and other organizations
 - General news outlets
 - Industry-specific news outlets
 - Social media
 - Miscellaneous sources available via search engines
- Proprietary third-party sources
 - Analyst reports and communications, including from "sell-side" or "Wall Street" analysts and credit rating agencies
 - Reports and data from platforms such as Bloomberg and FactSet
 - Reports and data from consultancies, often industry specific, such as Rystad in energy, IQVIA and Evaluate in biopharma, Gartner and IDC in information technology
- Proprietary primary research

- Surveys, conversations, product comparisons, and other studies commissioned by the analyst or conducted directly

Analysts in an institutional investment setting (e.g., an asset management firm) are often able to conduct this initial research quickly, as they enjoy broad access to a variety of proprietary third-party sources, may have industry knowledge and experience themselves, can access prior analyses of the company or industry, and often have access to the issuer investor relations personnel and management.

We will demonstrate the determination of an issuer's business model with a fictional example adapted from a real company. We will use this case study throughout this module and the next two to demonstrate the use of the company and industry analysis framework. Another case study of a different company in the same industry is in the self-assessment section at the end of each lesson, so you can practice using the framework on your own.

CASE STUDIES

Warehouse Club Inc.



Elaine Nguyen is an analyst at Fyleton Investments. To find investment candidates for one of Fyleton's funds, Nguyen ran a screen for companies in developed markets that exceeded a certain level of sales growth and return on invested capital over the last five years while also performing well in the last recession. One company that passed the screen is Warehouse Club Inc. ("Warehouse"). Neither Nguyen nor her colleagues have ever worked on Warehouse before. A portfolio manager suggested that Nguyen research the company and its industry for several days and then discuss initial findings with colleagues before writing a formal research report.

Warehouse is a public company, so Nguyen compiles several issuer information sources, including the company's latest annual report and a transcript of management's presentation at a recent investor event. The following paragraphs and figures are excerpts from these sources.

Warehouse Club Inc. ("we," "our," "us," "the company") is a leading discount retailer, consistently offering customers more than 20% savings on a broad range of food and other merchandise compared to supermarket, department store, convenience store, and e-commerce competitors. Warehouse operated 149, 145, and 141 stores as of 31 December 2X19, 2X18, and 2X17, respectively.

We sell a core assortment of packaged food and beverages, fresh foods, and non-food merchandise such as apparel, appliances, electronics and entertainment, housewares, and sporting goods. In addition to our core assortment, our stores feature a changing assortment of seasonal or discounted items based on the latest consumer trends and availability from suppliers, which vary by store. We also offer ancillary products and services such as petrol stations co-located with our stores, in-store takeaway prepared foods, pharmacy and optician centers, and hearing aids. In addition to carrying the leading branded goods, we also sell packaged food and household goods under our exclusive private-label brand that we source from contract manufacturers, which account for over 15% of our net sales and carry higher than average profitability.

We operate a member-based model that requires customers to show their membership card to enter and shop in our stores. The annual membership fee is \$60 per household. We estimate that members' annual savings versus shopping at competitors for the same basket of goods amounts to over five times their annual membership fee. Our membership fee revenues

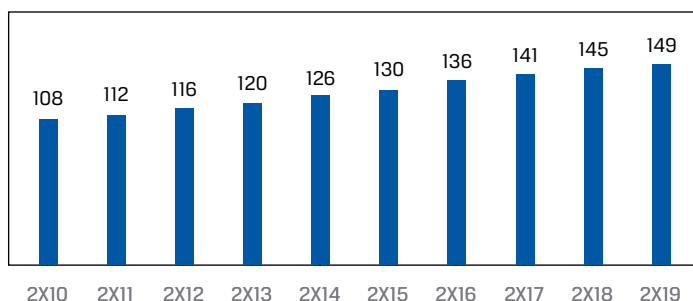
were \$601, \$576, and \$553 million in fiscal years 2X19, 2X18, and 2X17, respectively. Besides providing us with a source of recurring revenue, membership engenders shopper loyalty and drives our industry-low levels of inventory theft of less than 20 basis points as a percentage of net sales, which we believe is at least one-fifth that of competitors. Total memberships were 10.2, 9.8, and 9.4 million as of 31 December 2X19, 2X18, and 2X17, respectively.

Our strategy is based on price and cost leadership. We stock a limited selection of high-quality branded and private-label products in a wide range of categories that produce high sales volumes and rapid inventory turnover. We limit items to fast-selling models, sizes, and colors and carry an average of approximately 4,000 unique products per store, less than a quarter of the assortment of our supermarket and supercenter competitors. Most items are sold only in bulk sizes. Wherever possible, we utilize direct-from-manufacturer distribution and move merchandise immediately to the sales floor, presenting on pallets in a no-frills warehouse atmosphere. Our high-volume purchases of a limited number of products, direct from manufacturer distribution, and reduced handling of merchandise enable us to hold a low-cost position in the retail industry in terms of merchandise and labor costs. Our low costs and membership fee revenues enable us to operate profitably while charging significantly lower prices than other retailers. Low prices for quality products drive greater membership and net sales over time, leveraging our selling, general, and administrative expenses, reducing them as a percentage of net sales.

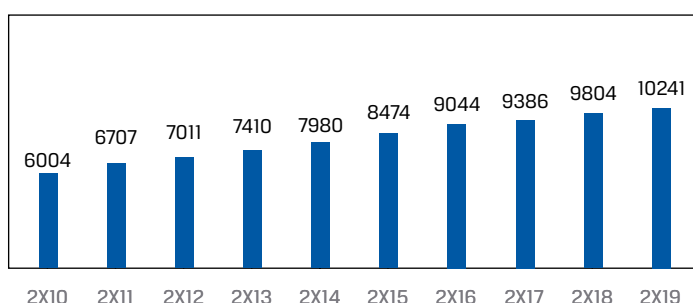
Our stores operate on a seven-day, 70-hour week with predictable shifts for employees. Because the hours of operation are shorter than other retailers, and due to other efficiencies inherent in a warehouse-type operation, labor costs are lower relative to the volume of sales. We look to maintain a large base of full-time employees with above-average wages and benefits to establish long tenures, as we believe this maximizes productivity and customer service at our stores. Our employee retention rate for employees who have been with us for at least a year is 90%, which we believe is substantially higher than the retail industry average.

Our retail operations, which represent substantially all our consolidated net sales, are our only reportable segment. We do not have significant sales outside our domestic geography: the United States. No single customer or supplier represents a material amount of revenues or merchandise costs, respectively.

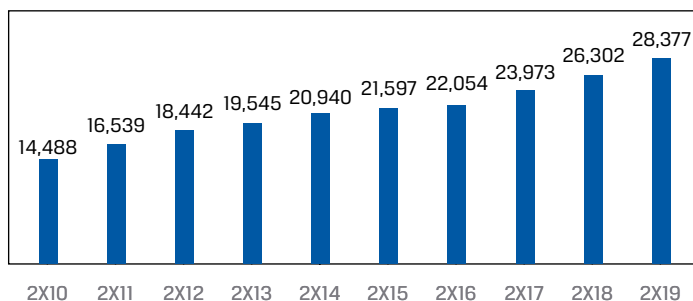
Warehouse Club Inc.
Number of Stores at Year-End, 2X10-2X19



Warehouse Club Inc.
Number of Members (thousands) at Year-End, 2X10-2X19



Warehouse Club Inc.
Annual Net Sales (USD millions), 2X10-2X19



Warehouse Club Inc. Income Statements for the Years Ended . . . (amounts in USD millions except per share amounts)

	31 December 2X17	31 December 2X18	31 December 2X19
Net sales	23,973	26,302	28,377
Membership fees	553	576	601
Total revenues	24,526	26,878	28,978
Merchandise costs	(21,258)	(23,399)	(25,248)
Selling, general, and administrative expenses	(2,201)	(2,363)	(2,566)
Depreciation and amortization	(260)	(273)	(283)
New-store opening expenses	(16)	(13)	(16)

	31 December 2X17	31 December 2X18	31 December 2X19
Operating income	791	830	865
Interest expense	(25)	(30)	(29)
Interest income	10	14	24
Income before taxes	776	814	860
Income tax expense	(178)	(187)	(198)
Net income	598	627	662
Shares outstanding	83	83	84
Earnings per share	7.20	7.55	7.88

During its presentation at a recent investor event, Warehouse management made the following statements that were not in the annual report:

- Three years ago, the company launched “Buy Online, Pick-Up in Store,” which enables members to shop on Warehouse’s website or mobile app and pick up their order, assembled by Warehouse employees, at a store. The company does not plan to offer e-commerce or delivery itself or through third-party services.
- Membership is \$60 per year, paid upfront, and refundable on a pro-rated basis for days remaining in the year. Management has increased the price of membership every five years by \$5 and expects to continue to do so. The last price increase was three years ago.
- Management intends to open four new stores per year for the next five years.
- The company operates stores in urban and suburban areas in a single country. Over 90% of current members live within 10 kilometers of a store, and while membership has grown around existing stores over time, management believes the primary driver of membership growth is opening stores in new areas.
- Management does not plan to open stores in another country for the foreseeable future. An immaterial amount of merchandise is imported.

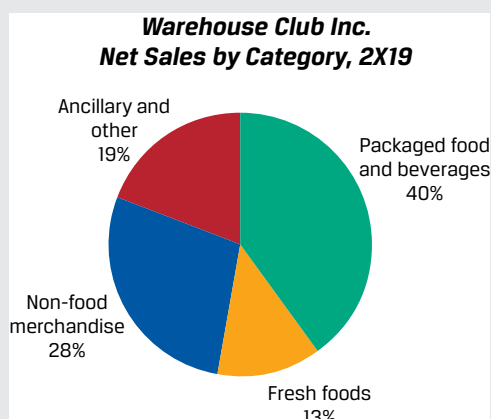
EXAMPLE 1

Determining Warehouse Club Inc.’s Business Model

Nguyen answers the following questions to determine Warehouse Club Inc.’s business model and to identify areas to conduct further research.

- What is the firm selling?

Warehouse sells a range of consumer goods, mostly consumables and perishables, approximately 4,000 unique items. For customers to shop at the stores, they must hold a membership, which Warehouse sells for \$60 per year, per household. Nguyen graphs the following net sales by category data with descriptions supplied by Warehouse, and notes that the composition of net sales by category has not changed materially in the last 10 years.



“Non-food merchandise” includes a broad range of items such as appliances, electronics, health and personal care products, hardware, outdoor and sporting goods, jewelry, furniture, and housewares.

“Ancillary and other” are service businesses that operate in or next to Warehouse’s stores and include takeaway prepared foods, prescription pharmacies, petrol stations, optician centers, and hearing aids.

■ To whom?

Warehouse’s members are primarily consumer households that live within 10 kilometers of a store, are value-conscious, are willing to forgo a more extensive selection for lower unit prices on bulk-sized goods, and are able to afford the annual membership fee.

■ How?

Warehouse sells products and services both in and around its stores. Members can either shop at the store or shop online and pick up at the store (Nguyen notes that management did not disclose the size of this in terms of net sales). Management said that it does minimal advertising (besides promotional mail) in an area before it opens a store and also sends out membership renewal reminders; the company primarily relies on word of mouth and its stores’ physical presence to drive interest and foot traffic.

Memberships can be purchased online, by phone, or at any Warehouse store. Management said that over three-quarters of members purchase and renew their memberships online.

■ For how much?

Warehouse has two revenue streams: goods and services (for which it earns net sales) and membership (for which it earns membership fees).

Management sets the prices of its merchandise by item. Gross margin and markup on aggregate net sales for the last three fiscal years were as follows:

	31 December 2X17	31 December 2X18	31 December 2X19
Net sales	23,973	26,302	28,377
Merchandise costs	(21,258)	(23,399)	(25,248)
Gross margin	2,715	2,903	3,129
Gross margin % of net sales	11.33%	11.04%	11.03%
Markup	12.77%	12.41%	12.39%

Therefore, the average good or service is sold at a price that is a little over 12% of the cost for Warehouse to purchase and ready the item for sale. While management claims that the company is a cost and price leader, Nguyen will have to corroborate that.

Membership is \$60 per year, per household. Membership is required to shop at Warehouse stores, thus effectively raising the price of all its goods and services.

- What does the company buy and rely on to operate effectively?
 - Merchandise from a broad range of manufacturers and private-label goods from contract manufacturers
 - Human capital in stores and in management
 - Fixed assets including land in attractive locations, buildings, fixtures, refrigerators and freezers, and a range of information technology systems such as point-of-sale payment systems
 - Credit and debit card payment networks and financial institutions to receive and make electronic payments

While these are all broadly available and the company has not had supply or service interruptions in the past, a key partner appears to be merchandise manufacturers, particularly those that manufacture Warehouse's private-label products. Since Warehouse is strict on cost and quality and carries a limited selection, the company does rely heavily on these contract manufacturers.

In summary, Warehouse's business model is aimed at creating a virtuous cycle whereby low prices drive memberships, which drive further sales, which drive negotiating leverage with suppliers to enable low prices at low costs.

DISCUSSION BOARD QUESTION



Based on the information provided, what would you change about or add to Elaine Nguyen's analysis of Warehouse Club Inc.'s business model? Why?

We encourage you to post your response, as well as read and reply to others' responses, on the Learning Ecosystem discussion board.

QUESTION SET



Iliso Marketplace Ltd. ("we," "our," "us," "the Company," "Iliso") is redefining e-commerce by delivering superior customer experiences at competitive prices through our Iliso Marketplace mobile app and website. While e-commerce customers have historically been forced to compromise on shipping times, product quality, and hassles with returns, we've built a vertically integrated system that delivers a vast selection of products with one-day delivery on most orders. We continue to make significant investments in both technology and our fulfillment infrastructure to improve the customer experience and expand into new product categories and geographies.

Customer Experience

We're committed to delighting our customers every day. Hallmarks of our customer experience include:

- **Next-Day or Faster Delivery.** Customers are eligible for one-day delivery 365 days a year. In 2X19, over 93% of orders were delivered within 24 hours of receiving the order.
- **Free Shipping.** Customers are eligible for free shipping on orders greater than \$25, which we believe is the industry-leading minimum order threshold, and no membership fee is required.
- **Vast Selection of Items, Including Fresh Groceries.** Customers can order from a selection of items across almost every category of goods—from fresh produce to consumer electronics to home décor and furniture. To continue offering the highest-quality items, we modify our selection continuously based on customer reviews. We enforce strict standards for third-party merchants to protect customers from counterfeit goods.
- **Frictionless Returns.** Our customers simply tap a button on our app and leave the item outside their door for pickup. Refunds are initiated the moment the item is picked up at the door.

The number of active customer accounts as of the fourth quarter for the years ended 31 December 2X17, 2X18, and 2X19, were 29, 31, and 36 million, respectively. Active customer account refers to a unique email address and at least one purchase in the indicated period; we do not charge membership fees or require minimum purchases to maintain an account.

Our Fulfillment Network

To achieve our differentiated customer experience, we focus on urban markets and have built a vertically integrated system in which we control every step, from a customer's purchase on our app to the photo confirmation of the delivery at the customer's front door. We receive most of our products directly from manufacturers at our fulfillment centers and minimize the use of third-party carriers and contractors by directly employing over 65,000 people in fulfillment and delivery. We believe direct employment and control of fulfillment assets such as delivery trucks enable speed, efficiency, and waste reduction, which is especially important for product categories such as fresh foods.

We've made the strategic decision to focus on urban markets in the United States for the foreseeable future to achieve an economical route density for customer deliveries and returns and one-day shipping.

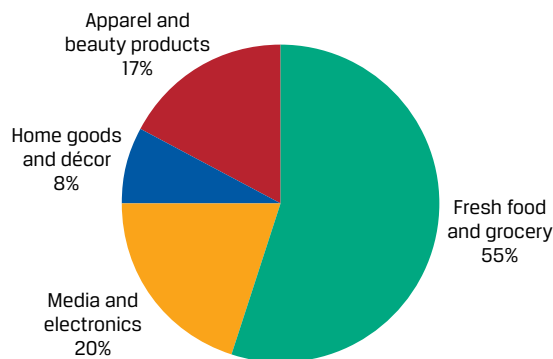
Merchant Services

In 2X13, we launched Merchant Services, allowing third-party merchants to sell products on Iliso Marketplace and utilize our fulfillment network. When multiple merchants (including us) sell the same product, our algorithm shows the customer the lowest price, including shipping cost, though the customer can manually choose a different seller. Third-party merchants must meet our rigorous quality standards but otherwise have discretion in product assortment and description, pricing, and shipping terms.

We receive sales commissions on third-party merchants' sales, as well as service revenues if the merchant utilizes our fulfillment network for warehousing and delivery. We record revenue on a "net" basis for third-party merchant-related transactions because we act as an agent; the Company is not the seller of record, nor does it take control of the goods inventory. The number of active third-party merchants as of the fourth quarter for the years ended 31 December 2X17, 2X18, and 2X19 was 25,500, 31,500, and 40,500, respectively. Active third-party merchant refers to a unique email address and at least one sale in the indicated period; we do not charge merchant membership fees, nor do we require minimum sales levels.

Our annual gross merchandise value (GMV)—which includes the value of all products sold on our marketplace by us and third-party merchants, net of sales taxes and returns allowances but including any applicable shipping fees—was \$5,721, \$6,950, and \$8,605 million for the years ended 31 December 2X17, 2X18, and 2X19, respectively. GMV by product category for the year ended 31 December 2X19 was as follows:

Iliso Marketplace Ltd.
GMV by Product Category, 2X19



Iliso recently went public, so financial information is available only for the five years ended 31 December 2X19. Amounts are in millions of USD (except per share amounts).

Iliso Marketplace Ltd. GMV, Customer, and Merchant Counts for the Years Ended 31 December

	2X15	2X16	2X17	2X18	2X19
Iliso retail sales	2,589	3,424	5,100	6,052	7,336
Third-party merchant sales	279	432	621	898	1,269
Total GMV	2,868	3,856	5,721	6,950	8,605
Active customer accounts (millions)	16	23	29	31	36
Third-party merchants (#)	14,850	18,000	25,500	31,500	40,500

Iliso Marketplace Ltd. Income Statements for the Years Ended 31 December

	2X15	2X16	2X17	2X18	2X19
Net retail sales	2,589	3,424	5,100	6,052	7,336
Third-party merchant fees	42	65	94	134	190
Total revenues	2,631	3,489	5,194	6,186	7,526
Costs of revenues	(1,894)	(2,506)	(3,729)	(4,446)	(5,414)
Operating, general, and administrative expenses	(685)	(895)	(1,300)	(1,544)	(1,831)
Advertising and marketing	(53)	(70)	(104)	(124)	(151)

	2X15	2X16	2X17	2X18	2X19
Operating income	(1)	18	61	72	130
Interest income	6	7	9	9	10
Interest expense	(5)	(13)	(15)	(12)	(8)
Pre-tax income	0	12	55	69	132
Provision for income taxes	0	(3)	(13)	(16)	(32)
Net income	0	9	42	52	100
Basic EPS	0.00	0.03	0.15	0.18	0.34
Diluted EPS	0.00	0.03	0.14	0.18	0.34
Basic share count	274	280	284	288	291
Diluted share count	274	280	291	288	297

The notes to Iliso's financial statements provide the following descriptions of accounts.

Net retail sales

Retail sales are earned from the Company's online product sales to consumers. Retail revenue is recognized when control of the goods is transferred to the customer, which occurs upon delivery to the customer.

Third-party merchant fees

Third-party merchant fees revenue includes commissions and service fees earned from merchants that sell their products through the Company's online business. The revenue is recognized when the order is completed and transmitted to the third-party merchant.

Costs of revenues

Costs of revenues primarily comprise the purchase price of products sold to customers where the Company records revenue gross, and include logistics center costs. Inbound shipping and handling costs to receive products from suppliers are included in inventory and recognized in costs of revenues as products are sold. Additionally, costs of revenues include outbound shipping and logistics-related expenses, delivery service costs from our Merchant Services business (where the Company is the delivery service provider), and depreciation expense.

Operating, general, and administrative expenses

Operating, general, and administrative expenses include costs incurred in operating and staffing the Company's fulfillment centers (including costs attributable to receiving, inspecting, picking, packaging, and preparing customer orders), customer service costs, payment processing fees, costs related to the design, execution, and maintenance of the Company's technology infrastructure and online offerings, and general corporate function costs.

Advertising and marketing expenses

Advertising and marketing expenses are expensed as incurred and include brand and performance advertising such as television and digital advertising, sponsored search, email marketing campaigns, and other similar initiatives.

1. What forms the customer base (the "who") for Iliso's merchant services segment?

- A. Third-party merchants
- B. E-commerce customers

C. Fresh food and grocery consumers

Solution:

A is correct. In merchant services, Iliso acts as an agent for third-party merchants. Iliso is paid commissions and service fees by merchants that sell their products through the company's online business. Third-party merchants are therefore the customer base for this business segment.

B is incorrect, because e-commerce customers are customers of Iliso's retail sales segment. Iliso is paid by e-commerce customers for goods purchased through its online platform. The purchases of third-party merchant products by these e-commerce customers belong to the third-party merchant and are not recognized in Iliso's retail sales.

C is incorrect, because fresh food and grocery consumers are a product line segmentation of both its retail sales and its third-party merchant GMV. In addition, fresh food and grocery forms only 55% of the overall GMV and does not encompass all of Iliso's GMV sales.

2. Iliso's business model(s) is(are) likely to benefit from _____ effects.

- A.** network
- B.** value chain
- C.** supply chain

Solution:

A is correct. Iliso's business model—in particular, merchant services—is an example of a two-sided network effect. Network effect refers to a network in which the value increases as more participants join the network. By providing an excellent service to merchants and to consumers, Iliso is able to attract both buyers and sellers to its online marketplace. The growth of users on Iliso's platform attracts more merchants, which in turn attracts more users.

B is incorrect, because value chain refers to the systems and processes within a firm that create value for its customers—which answers the “how” aspect of a business model.

C is incorrect, because the supply chain is the network inside and outside the company involved in producing the product/service and delivering it to the end user.

3. Which of the following business features is *least likely* to increase consumer retention (“stickiness”) on Iliso's platform?

- A.** A focus on urban markets
- B.** Next-day or faster delivery
- C.** No membership fee is required.

Solution:

A is correct. A focus on urban markets does not change customer stickiness. The customer experience for those in urban markets would not necessarily change if the company began shipping to rural markets as well.

B is incorrect. The company states that next-day or faster delivery is a hallmark of its customer experience and may provide a competitive advantage.

C is incorrect, because the lack of a membership fee is likely to decrease the cost of a customer's staying on the platform, thus increasing retention for some customers.

4. The gross margin of Iliso's merchant services segment is *most likely* _____ the gross margin of Iliso's retail sales segment.

- A. higher than
- B. the same as
- C. lower than

Solution:

A is correct. The cost of revenues for Iliso's retail sales includes the purchase price of products sold to customers, including shipping and handling costs, whereas the cost of revenues for merchant services includes only the out-bound shipping and logistics expenses. The purchase price of products is a far higher proportion of total GMV sold to customers compared with shipping and logistics costs. Accordingly, the exclusion of the purchase price of products in both the numerator and the denominator will raise the reported gross margin of the merchant services segment.

B is incorrect, because the gross margin for merchant services cannot be the same as that of retail sales. The retail sales segment includes the purchase price of products sold and associated inbound logistics costs, which means the numerator and the denominator are significantly higher than in the merchant services segment.

C is incorrect, because the gross margin for merchant services cannot be lower than that of retail sales. Assuming a positive gross margin for both segments, the gross margin for merchant services must be higher since the denominator is smaller (being only the service commissions and service fees charged to merchants).

REVENUE ANALYSIS

4



evaluate a company's revenue and revenue drivers, including pricing power

Revenue Drivers

After determining the business model, we turn to the analysis of historical financial results and position for the company. This analysis will be used as a basis for forecasting a financial statement model that supports security valuation. For most companies—perhaps with the exception of banks, for which the balance sheet comes first—analysts start by analyzing revenues. As with other financial statement lines, the analysis involves identifying **drivers**, which are causative factors that explain the level of and changes in an output variable (here, revenues), and understanding the evolution of the drivers over time.

Analysts can take a bottom-up or top-down approach to determining revenue drivers. A bottom-up approach decomposes revenues into drivers such as sales volume and price, or revenues by product line, segment, or geography, which may be further broken down into other drivers. A top-down approach expresses revenues as a function of drivers such as market share, the addressable market or market size, and GDP growth. The two approaches are often used together.

CASE STUDIES**Bottom-Up Approach to Revenue Analysis:
Warehouse Club Inc.**

Warehouse's revenues are composed of net sales and membership fees.

Based on her business model work, Nguyen knows that net sales are sales of merchandise and services in Warehouse stores (the company does not operate an e-commerce business). A logical driver of net sales each year, therefore, is the number of stores open that year. Nguyen decomposes Warehouse's net sales into two drivers: the *average* number of stores open and net sales per average store. Nguyen also calculates the change in these drivers: the absolute change in stores each year (new-store openings; so far, Warehouse has never closed a store) and the annual percentage change in net sales per store.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Stores, year-end	108	112	116	120	126	130	136	141	145	149
New-store openings	5	4	4	4	6	4	6	5	4	4
Average stores (#)	106	110	114	118	123	128	133	139	143	147
Net sales per average store	138	150	162	166	170	169	166	173	184	193
Growth rate		9%	8%	2%	3%	-1%	-2%	4%	6%	5%
Net sales	14,488	16,539	18,442	19,545	20,940	21,597	22,054	23,973	26,302	28,377

Net sales in 2X19 were 96% higher than in 2X10, driven by similar growth contributions from the two drivers: the average number of stores increased by 39% and sales per average store increased by 40%.

An encouraging sign for forecasting this company's results is that each driver appears relatively stable, indicating that they might be somewhat predictable. Sometimes, however, key revenue drivers are volatile, such as interest rates and oil prices for banks and oil producers, respectively. Volatility does not mean that the analyst has selected an incorrect driver; drivers are based on the business model.

Nguyen then decomposes Warehouse's membership fee revenues into the annual price of membership and the average number of members.

Members

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Members, end of year	6,004	6,707	7,011	7,410	7,980	8,474	9,044	9,386	9,804	10,241
Members, annual average	5,900	6,360	6,855	7,218	7,691	8,218	8,764	9,217	9,600	10,017
Avg. per member fee	50.00	50.00	55.00	55.00	55.00	55.00	55.00	60.00	60.00	60.00
Membership fee revenues	295	318	377	397	423	452	482	553	576	601

Membership growth has been a larger driver of membership fee revenue growth than price increases, as the number of members has grown by 70% from 2X10 to 2X19 while prices have increased by 20% over the same period.

Management indicated that over 90% of members live within 10 kilometers of a store and, from 2X10 to 2X19, the company opened 46 new stores. Nguyen adds to her membership analysis by calculating the average number of members per store; if this number has been falling over time, it could mean that existing stores are losing members and sales to new stores the company opens, an example of **cannibalization**.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Avg. members per store	57,282	57,818	60,128	61,171	62,528	64,205	65,892	66,546	67,133	68,141
Growth rate		0.9%	4.0%	1.7%	2.2%	2.7%	2.6%	1.0%	0.9%	1.5%

Not only has the average number of members per store not declined, but it has also grown by 19%, indicating no cannibalization and that new-store openings are not the only driver of membership growth. This is a positive sign, though Nguyen notes that additional analysis is needed to estimate a ceiling on the number of members a store can reasonably support, and to determine whether stores that opened many years ago have already hit this ceiling.

Pricing Power

Prices are a driver of revenue for every firm, and it may appear that pricing is a unilateral decision by management (e.g., Warehouse management chooses its markups on merchandise net sales and has chosen to increase its annual membership fee by \$5 every five years). In fact, prices are constrained by a company's **pricing power**. Pricing power refers to a company's ability to set prices and other economic terms with customers without affecting its sales volumes. While Warehouse's management could choose to increase the price of membership from \$60 to \$600, such a decision would likely result in the loss of all its members. The membership price is constrained by the savings Warehouse can offer members on purchases, which itself is the subject of competition from other retailers that also use low prices to attract customers and of negotiations with suppliers that desire higher prices. Pricing power is primarily a function of both market structure (covered earlier in Economics) and a company's competitive positioning in its market.

In the most competitive markets, where firms are selling nearly identical products, firms are price *takers*—that is, price is dictated by the forces of supply and demand—and all firms generally sell at the same price. If a firm prices above the prevailing market price, it will sell a fraction or none of its output, as customers will buy elsewhere. It is also generally not rational for a firm to price below the market price, as doing so would not only reduce its profits but also cause competitors to do the same, leading to a price war. Returns on capital in highly competitive markets move in cycles based on supply and demand, but over the long run approximate firms' cost of capital as competition forces prices down to marginal cost. A **low-cost producer** (with costs below—in some cases, *well* below—those of a marginal producer) can, however, earn long-run returns above its cost of capital in a highly competitive industry, but it requires the low-cost position to be sustained against competition over time. A prominent example is state-owned oil producers that benefit from monopoly ownership of low-cost oil reserves in certain countries.

Other attributes of highly competitive markets include little to no product differentiation, low barriers to firm entry, available substitutes, a lack of customer loyalty, and low switching costs for customers. Many markets fit this description, including retail, oil and gas and other natural resources, bank loans and deposits, insurance, fresh food, off-patent pharmaceuticals, transportation, and some types of technology

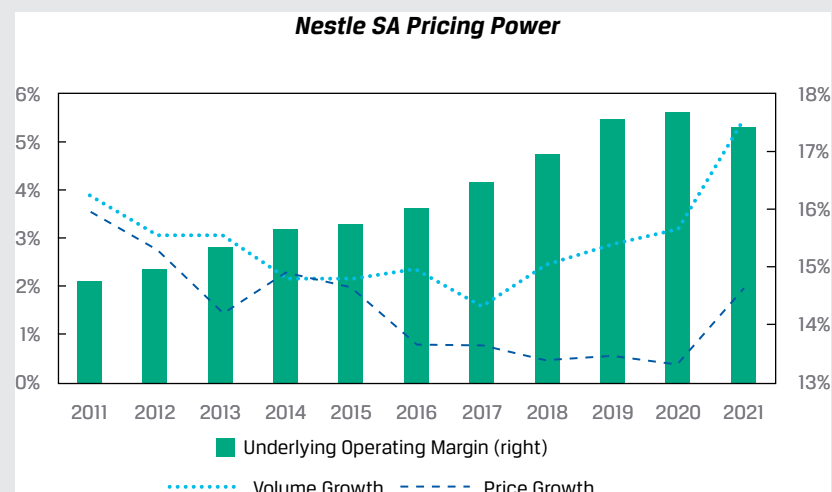
hardware. Highly competitive markets often do not start as highly competitive but, rather, become more competitive as new firms enter, the pace of innovation slows, and imitation becomes widespread, a process known as **commoditization**.

Firms operating in less competitive markets, such as monopolistic competition, oligopoly, or monopoly structures, tend to have some degree of pricing power. They can raise prices or change other economic terms (contract length, late fees, etc.) without materially affecting sales volumes. Attributes of these markets include product differentiation, barriers to entry, lack of economical substitutes, switching costs for customers, and a high degree of customer loyalty or retention. Markets and companies that fit this description include branded consumer goods, utilities, patented pharmaceuticals, software, restaurants, payment networks, medical devices, and some types of real estate. Firms with pricing power often apply value- or cost-based pricing as well as price discrimination strategies (outlined in the previous module). While product differentiation is most visible, differentiation can also include convenience, wrap-around services, and a secondary resale market.

Analysts evaluate pricing power not just by examining a firm's prices over time or relative to competitors, but also by comparing a firm's prices with its costs—in other words, its profit margins. While a company may be increasing prices by 3% per year, if its costs are rising by 5%, it is showing both an inability to pass along price increases to customers and declining profitability for investors. A company without pricing power may be unable to increase prices even as costs rise, because it is constrained by the availability of substitutes and customers can easily switch. For example, a movie theater might be unable to increase prices in line with costs, because streaming video subscriptions are cheaper and the company would likely lose customers if it raised prices significantly. Therefore, an important sign of pricing power is rising profitability over time, which demonstrates both an ability to retain economics for investors and that competitors, new entrants, or substitutes are not driving down prices faster than costs.

EXAMPLE 2

Nestlé SA is a global packaged food and beverage company headquartered in Switzerland. The company sells products with proprietary brand trademarks (such as Nespresso, Purina, Perrier, Maggi, and Stouffer's) that it acquires and develops with marketing and R&D investments, which totaled CHF19 billion in 2021, or 20% of the company's revenues. The distinctiveness of its products through branding can confer pricing power. From 2011 to 2021, the company demonstrated its pricing power through a record of increasing prices while increasing volumes and its operating profitability.



Top-Down Revenue Analysis

Analysts can also take a top-down approach to analyzing revenue by expressing it as a function of drivers such as market size and a company's share of that market. This approach will be discussed in more detail in the next module (on industry and competitive analysis), but we will demonstrate this approach for Warehouse Club Inc.

CASE STUDIES

Top-Down Approach to Revenue Analysis: Warehouse Club Inc.



A government agency reports total national retail and food services sales each month for the country that Warehouse operates in (the United States). The data capture consumer spending at hundreds of thousands of retailers and restaurants. Sales are categorized by the type of retailer or restaurant, which reflects the primary type of product or service it sells. An excerpt of a monthly release is shown below.

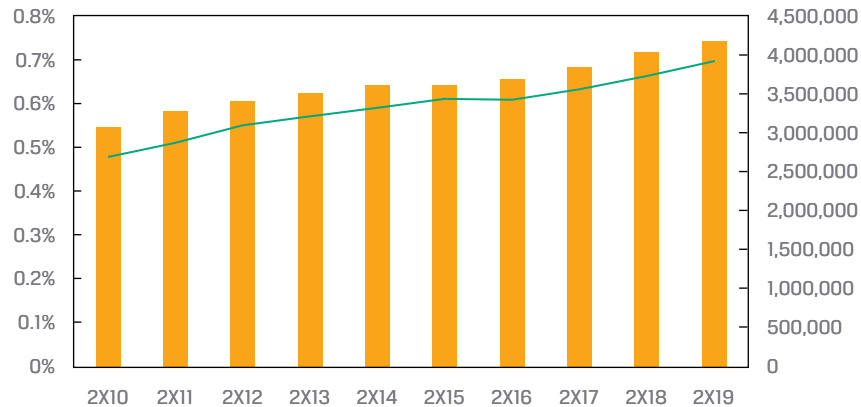
National Retail Sales, January 2X19 (billions of USD)	
Total retail and food services sales	456,012
Total retail sales	399,359
Automobile and parts dealers	88,152
Petrol stations	36,539
Pharmacies and drug stores	24,273
Grocery stores	57,392
Total food services sales	56,653
Full-service restaurants	24,730
Limited-service restaurants	25,273
Pubs, bars, and other	6,650

Warehouse sells a broad range of retail goods in most cities, so US national retail spending is generally representative of Warehouse's **market size**—the existing demand for goods or services offered by a company. By expressing Warehouse's revenues as a percentage of the market, or its **market share**, over time, we can assess whether the company has been gaining or losing relevance with its customers relative to competitors and substitutes.

A challenge for analysts is determining market size. The data might be difficult to gather (not the case here), and judgment is required for determining what to include and exclude: only sales of products identical to the company's, sales of similar products, all sales of competitors, and so on. The common practice is to include all sales of *similar* products and to consider *substitutes* separately (discussed in the next module on industry analysis). A substitute serves the same function as a product but differs in form (e.g., movie theater and streaming video, grocery stores and restaurants, print and digital advertising). This practice is subject to a wide range of opinions, however, and some analysts include sales of substitutes in market size estimates. The share of market that a firm does not have (i.e., $100\% - x\%$ market share) represents sales potential.

Elaine Nguyen decides to use total US retail sales as the market size for Warehouse, with one exclusion: automobiles and parts. Warehouse does not sell automobiles or automobile parts, nor does it sell substitute means of transportation or repair. Nguyen considers food service substitutes (restaurants and pubs) for Warehouse.

Warehouse Club Inc. Market Share (% left) of Retail Sales excl. Autos and Parts (USD millions, right)



Nguyen makes three observations:

- The market is large and growth has been positive, albeit at a slow (3.4%) rate. This finding aligns with Nguyen's expectations, because this is a developed market that represents a broad range of consumer spending (a large portion of which is for necessities) and the country has been in an economic expansion.
- Warehouse Club Inc. has a small (70 basis points in 2X19) but growing share of the market (up from 48 basis points in 2X10), indicating that its value proposition relative to other retailers has been resonating with consumers.
- Warehouse's market share increased in most years, with the exception of 2X16, when its share was flat from the prior year. Nguyen will have to investigate further.

DISCUSSION BOARD QUESTION



Which of Nguyen's analyses was more helpful to you in understanding Warehouse's historical revenue growth: the bottom-up or the top-down approach? Why?

We encourage you to post your response, as well as read and reply to others' responses, on the Learning Ecosystem discussion board.

QUESTION SET



Questions 1–4 refer to the earlier vignette on Iliso Marketplace Ltd.

1. In 2X19, the percentage of GMV that Iliso collected from third-party merchants (the company's "take rate") was *closest* to:

- A. 2.2%.
- B. 14.7%.
- C. 15.0%.

Solution:

C is correct. For e-commerce businesses, the take rate refers to the retention percentage of the value of transactions facilitated by the platform. In 2X19, Iliso third-party merchant fees were \$190 million against third-party gross merchandise value (GMV) of \$1,269 million on its platform. The take rate is therefore:

$$190/1,269 = 0.1497 \approx 15.0\%$$

A is incorrect, because it results from dividing third-party merchant fees by total GMV. B is incorrect, because 14.7% is the percentage of Iliso's GMV attributable to third-party merchant sales.

2. Which of the following measures would be used to best assess Iliso's pricing power?

- A. Take rate
- B. GMV year-on-year growth rate
- C. GMV per active customer account

Solution:

A is correct. The percentage of transaction value retained by the platform reflects the price Iliso charges third-party merchants to use its platform. Both the stability and the level of its take rate reflect the state of competition that Iliso faces from other e-commerce platforms.

B is incorrect, because the GMV year-on-year growth rate can be affected by numerous factors, including customer volumes, product mix, and basket size. It does not carry information for analysts to estimate pricing power.

C is incorrect, because GMV per active customer account can be affected by product mix and the average basket size, which is driven by its \$25 minimum for free shipping. This measure does not carry information for analysts to estimate the pricing power of the platform.

3. Referring to the data table earlier in this lesson titled "National Retail Sales, January 2X19" and assuming 2X19 US monthly retail and food sales of \$456 billion and assuming Iliso's addressable geography is the United States, Iliso's market share of retail and food sales was *closest* to:

- A. 0.134%.
- B. 0.138%.
- C. 0.157%.

Solution:

C is correct. In 2X19, annual US retail and food sales were: $456 \times 12 = \$5,472$ billion. Since GMV reflects the value of retail sales on Iliso's platform, the appropriate estimate of Iliso's market share is therefore:

$$2X19 \text{ GMV} / 2X19 \text{ US retail and food sales} = 8,605 / 5,472,000 = 0.157\%$$

A is incorrect, because this measure uses only Iliso's retail sales and ignores third-party GMV, even though the vignette states that merchants may sell the same products.

B is incorrect, because this measure uses Iliso's net revenues as a percentage of the market. This measure is inappropriate, since Iliso's net revenues do not reflect the gross value of third-party merchant sales on its platform.

4. Which of the following is an appropriate bottom-up driver of Iliso's revenues?

- A. Iliso's market share of US retail and food sales
- B. GMV per average active customer account
- C. Iliso's revenue as a function of GDP growth

Solution:

B is correct. In a bottom-up approach, analysts decompose the revenues into logical revenue drivers such as product line or segment. A driver of Iliso's net sales must be the average number of active customer accounts during the period and the net retail sales per account. We use GMV, since this measure accurately reflects the total amount of retail sales (both Iliso's first-party and third-party sales).

A and C are incorrect, because they are top-down, not bottom-up, drivers.

5

OPERATING PROFITABILITY AND WORKING CAPITAL ANALYSIS



evaluate a company's operating profitability and working capital using key measures

Operating Costs and Their Classification

Generally, operating costs are incurred in generating—or are otherwise related to—*current period* revenue: all costs related to the acquisition, production, sale, improvement, and delivery of goods and services; the management of business activities; and compliance with laws and regulations. Investing costs are related to the acquisition and production of long-term assets, including tangible assets like property and equipment and intangible assets such as software, trademarks, and patents. Financing costs include payments to debt and equity investors as a return on their investment.

MINDING THE GAAP: CLASSIFICATION OF COSTS

IFRS and US GAAP do not classify operating, investing, or financing costs in an intuitive manner, and there are differences in classification on the income statement and statement of cash flows. Several prominent examples include:

Cost	Intuition	IFRS and US GAAP Classification: Income Statement	IFRS and US GAAP Classification: Statement of Cash Flows
Research and development	Investing	Operating	Operating
Depreciation and amortization*	Investing	Operating	Non-cash add-back to Operating <i>Capital expenditures</i> in Investing
Interest expense	Financing	Other (not operating)	US GAAP: Operating IFRS: Operating or Financing
Income taxes	Separate	Other (not operating)	Operating

Recall that IFRS and US GAAP require the costs of acquiring and constructing long-term tangible and intangible assets to be capitalized and then, for those with a finite useful life, systematically expensed over time on the income statement as non-cash depreciation and amortization expense. The most common allocation method is straight line, where the cost is expensed evenly over the useful life of the asset.

Unless specified otherwise, we use the IFRS classifications of costs on the income statement in our discussion.

For most companies, operating costs account for the majority of costs and are primarily determined by the company's business model and size. As shown in Exhibit 4, a company's operating costs can be categorized and analyzed in three ways: by their behavior with output, their nature, or their function.

Exhibit 4: Categorization of Operating Costs

Behavior with Output <i>Whether the cost varies or not with output in the short run</i>	Nature <i>What the cost is</i>	Function <i>The purpose of the cost</i>
- Variable costs vary with output - Fixed costs do not vary with output	- Compensation of employees - Raw materials - Merchandise - Office supplies	- Cost of goods sold - Sales and Marketing - General and administrative - Research and development

Behavior with Output: Fixed and Variable Costs

A useful classification of operating costs for analysts is fixed versus variable, which was introduced in an earlier module. The distinction is useful because the proportion of fixed to variable costs in operating costs affects the stability and predictability of operating profit. Using a fixed/variable operating cost classification, operating profit is defined in Equation 1.

$$\text{Operating profit} = [Q \times (P - VC)] - FC \quad (1)$$

Q = units of outputs sold in a period.

P = price per unit of output.

VC = variable operating costs expressed per unit of output. Examples include merchandise costs for a retailer like Warehouse Club Inc. and materials and direct labor costs for a manufacturer.

FC = fixed operating costs, which do not change within a given range of output in the short run. Examples include compensation for salaried employees, depreciation and amortization, software and IT expenses, insurance, and certain utilities costs. FC is stated on a total dollar, not per unit, basis.

For a firm to be profitable, it's clear from Equation 1 that $(P - VC)$, or the **contribution margin**, must be positive and that Q must be high enough such that FC is exceeded. Recall that the amount of fixed costs in the operating cost structure of an issuer is referred to as operating leverage and that it presents both benefits and risks. If operating costs are largely fixed and the contribution margin is positive, operating profit can increase rapidly with increases in Q . However, the reverse is true if Q declines, since fixed costs do not change, so operating profit will fall. As shown in Equation 2, operating leverage can be measured and compared across firms by using the **degree of operating leverage** (DOL):

$$DOL = \% \Delta \text{ Operating Profit} / \% \Delta \text{ Sales} \quad (2)$$

A firm can increase its degree of operating leverage by increasing the fixed costs and decreasing the variable costs in its cost base.

EXAMPLE 3

Variable and Fixed Cost Analysis

While analyzing operating costs using the variable and fixed framework is insightful, its use by independent investment analysts is limited because disclosure of fixed and variable costs is not required by IFRS or US GAAP, nor is the disclosure of output volume. Additionally, many companies have several lines of business that have different volume, price, and cost structures, the details of which are lost in the presentation of highly aggregated financial statements that investors receive. However, in some industries, it is common for issuers to disclose these figures, or it is relatively easy for an analyst to make assumptions. A prominent example is oil and gas producers, for which we provide an example here.

Ribbon Energy Ltd. ("Ribbon") is a fictional oil and natural gas company focused on the exploration and development of oil and natural gas reserves. Selected operating and financial data for the last 12 months are as follows:

	Last 12 Months
Sales volume (thousands of barrels of oil equivalent)	154,812
Average price, per barrel of oil equivalent	\$49.66
Total revenues (millions)	\$7,688
<i>Production costs, per barrel of oil equivalent:</i>	
Lease operating expense	11.21
Production and ad valorem taxes	3.40

	Last 12 Months
Gathering and transportation expense	2.63
Production costs	\$17.24
<i>Other operating expenses (millions of USD):</i>	
Depreciation, depletion, and accretion (DD&A)	1,275
General and administrative expenses (G&A)	150

Production costs are variable costs stated on a per barrel of oil equivalent basis, while other operating expenses (DD&A and G&A) are fixed and stated on an aggregate dollar basis.

Ribbon's management provides the following guidance for the next 12 months. Management does not provide guidance for prices (and therefore revenue), since it sells in a commodity market with volatile prices that are difficult to predict. Ribbon does not hedge. At the time guidance was provided, the average oil price per barrel was \$62.50.

	Next 12 Months
Sales volume (thousands of barrels of oil equivalent)	156,360–167,197
Production costs, per barrel of oil equivalent	\$17.34
<i>Other operating expenses (millions of USD):</i>	
Depreciation, depletion, and accretion (DD&A)	1,415
General and administrative expenses (G&A)	150

1. Ribbon's contribution margin per barrel for the last 12 months is *closest* to:

- A. \$23.2.
- B. \$32.4.
- C. \$45.3.

Solution:

B is correct. Recall that the contribution margin per barrel is $P - VC$. The average price per barrel, P , in the past 12 months was \$49.66, and the variable cost per barrel, VC , was \$17.24. Therefore, the contribution margin was $\$49.66 - \$17.24 = \$32.42$.

A is incorrect, because this amount is the company's EBIT, or operating profit, per barrel.

C is incorrect, because it results from using the current rather than the last 12 months' average oil price per barrel to calculate the contribution margin.

2. At the high end of management's guidance, Ribbon's operating profit for the next 12 months is *closest* to:

- A. \$5,496 million.
- B. \$5,986 million.
- C. \$6,002 million.

Solution:

B is correct.

$$\text{Operating profit} = [Q \times (P - VC)] - FC$$

$$\text{Operating profit} = [167.197 \text{ million} \times (\$62.50 - \$17.34)] - (\$1,415 + \$150)$$

$$\text{Operating profit} = \$5,986 \text{ million}$$

A is incorrect, because this amount is the operating profit using the low end of guided sales volume of 156.360 million barrels.

C is incorrect, because it relies on last year's production costs of \$17.24 per barrel.

3. Some analysts covering Ribbon believe that volume will be lower than what management expects. Assuming a 5% reduction over last year's volume, Ribbon's operating profit for the next 12 months is *closest* to:

- A. \$5,077 million.
- B. \$5,091 million.
- C. \$5,776 million.

Solution:

A is correct.

The last 12 months' volume was 154.812 million. A 5% reduction is $154.812 \times 0.95 = 147,071$ million barrels.

$$\text{Operating profit} = [Q \times (P - VC)] - FC$$

$$\text{Operating profit} = [147,071 \text{ million} \times (\$62.50 - \$17.34)] - (\$1,415 + \$150)$$

$$\text{Operating profit} = \$5,076.7 \text{ million}$$

B is incorrect, because it uses last year's production costs.

C is incorrect, because it assumes a 5% increase rather than decrease in volume from the prior year.

4. Ribbon's degree of operating leverage for the next 12 months at the low end of management's guidance is _____ at the high end of management's guidance.

- A. lower than
- B. the same as
- C. higher than

Solution:

C is correct.

Last 12 months' sales: \$7,688

Last 12 months' operating profit: \$1,244

Low end of guidance

Next 12 months' sales: $156.360 \times \$62.50 = \$9,773$

Next 12 months' operating profit: $\$9,773 - (156.360 \times 17.34) - 1,565 = 5,496$

Degree of operating leverage: $(5,496/1,244 - 1)/(9,773/7,688 - 1) = 1.95$

High end of guidance

Next 12 months' sales: $167.197 \times \$62.50 = \$10,450$

Next 12 months' operating profit: $\$10,450 - (167.197 \times 17.34) - 1,565 = 5,986$

Degree of operating leverage: $(5,986/1,244 - 1)/(10,450/7,688 - 1) = 1.85$

Natural and Functional Operating Cost Classifications and Measures of Operating Profitability

Rather than fixed and variable, IFRS and US GAAP require issuers to disclose operating costs using either a natural or a functional cost classification based on “historical and industry factors and the nature of the entity.” Most issuers, including Warehouse Club Inc., choose a functional classification. This approach leads to all issuers’ income statements appearing similar in structure, with lines related to functional areas such as “cost of sales,” “selling, general, and administrative expenses,” and “research and development,” even though their business models (and fixed versus variable cost composition) can differ significantly. The functional classification of operating costs is used to calculate and distinguish key profitability measures used in analysis and forecasting, as shown in Exhibit 5.

Exhibit 5: Measures of Operating Profitability

Gross Profit:	EBITDA:	Operating Profit:
Revenue	Revenue	Revenue
Cost of Sales	Cost of Sales	Cost of Sales
Gross Profit	Operating Expenses i.e. SG&A, R&D, etc.	Operating Expenses i.e. SG&A, R&D, etc.
	EBITDA	Depreciation and Amortization
		Operating Profit (EBIT)
Gross Margin:	EBITDA Margin:	Operating/EBIT Margin:
Gross Profit	EBITDA	Operating Profit (EBIT)
Revenue	Revenue	Revenue

Gross profit, EBITDA, and EBIT are measures of operating profitability that capture different functional costs in their definitions. Generally, a significant amount of cost of sales is variable, so gross margin is an approximate measure of contribution margin. Other operating expenses such as sales and marketing, general and administrative, and research and development costs tend to be largely fixed, though they may have a variable component such as sales commissions. Depreciation and amortization expenses are fixed unless the issuer uses units of output as the depreciation method. An issuer’s notes to its financial statements contain useful information about the composition and nature of operating costs.

For most companies, the major driver of operating costs over the long run is output, even for fixed costs, because output growth often requires growth in assets, human capital, and purchased goods and services. Warehouse Club Inc., for example, might be able to grow membership fee revenues to an extent without incurring costs, but growing net sales requires more stores and employees. This has been true even for so-called asset-light companies in the internet and software industries, which have

shown large increases in operating costs along with large increases in revenue over time. Since output or revenue is a major cost driver, analysts often express operating costs as a percentage of revenue. Additional considerations include industry profitability, **economies of scale**, and **economies of scope**.

Industry profitability will be discussed in detail in the next module, but recall from the earlier discussion on pricing power that prices relative to costs (i.e., profitability) are largely dictated by market structure and a company's competitive positioning. Since industry participants are competing in the same product and factor markets, competitive forces within them determine industry profitability in the long run; analysts should not analyze profitability only on a company-by-company basis.

Economies of scale refer to a decline in costs per unit as output grows and generally result from having fixed costs in the cost structure that are spread over more units of output. A company with entirely variable costs can also exhibit some economies of scale over time if it increases its bargaining power over suppliers as it grows, driving down variable costs per unit. Economies of scale are usually evident from the business model and can be found empirically by comparing a company's size in revenues with its operating costs as percentages of sales or margins.

Economies of scope refer to a decline in costs per unit as the number of product or business lines increases and generally result from shared costs between the product lines. Economies of scope are usually evident from the business model and can be found empirically by comparing the profitability of an integrated company with that of a standalone company. Examples of economies of scope can be found in financial services, where there are customer, client service, compliance, technology, and back-office similarities (and thus cost efficiencies) across commercial banking, consumer banking, brokerage, asset management, payment processing, and investment banking. Economies of scope have resulted in large global firms competing in many lines of business.

Recall that Warehouse Club Inc.'s management claimed that it has economies of scale, saying that "low prices for quality products drive greater membership and net sales over time, leveraging our selling, general, and administrative expenses, reducing them as a percentage of net sales." While this makes intuitive sense as there is probably a fixed amount of SG&A costs each year to operate a store, Elaine Nguyen must corroborate it empirically.

CASE STUDIES

Warehouse Club Inc. Operating Profitability Analysis



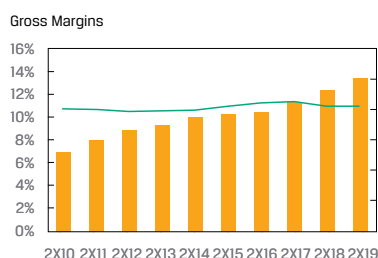
Warehouse Club Inc.'s operating costs are composed of merchandise costs (costs of sales), SG&A expenses, depreciation and amortization, and costs associated with opening new stores.

Based on her business model work, Elaine Nguyen believes the following about drivers for each of these operating costs.

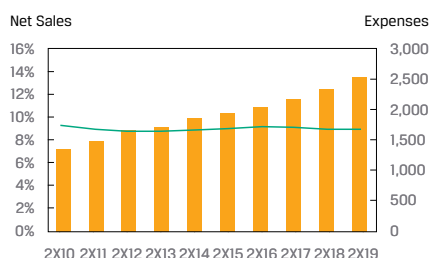
Operating Cost	Driver(s)
Merchandise costs (costs of sales)	Sales
SG&A expenses	Sales Number of employees and per-employee compensation
Depreciation and amortization	Long-term assets Useful lives
Store opening expenses	Store openings

Based on these drivers, Nguyen calculates Warehouse's gross margin (1 – cost of sales as % of net sales), SG&A expenses as a percentage of net sales, gross fixed assets to depreciation and amortization (implied useful life), store opening expenses per new-store opening, and, finally, profit margins. Profit margins include membership fee revenues.

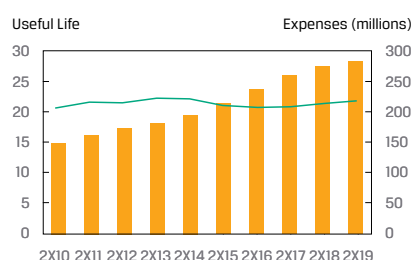
Warehouse Club Inc. Gross Margin (line, left) and Cost of Sales (bars, right, USD millions)



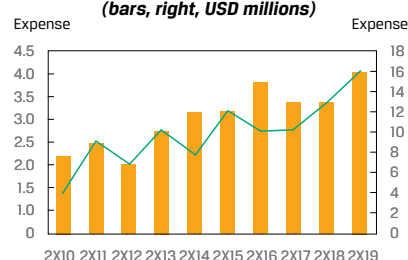
Warehouse Club Inc. SG&A % of Net Sales (line, left) & SG&A Expenses (bars, right, USD millions)



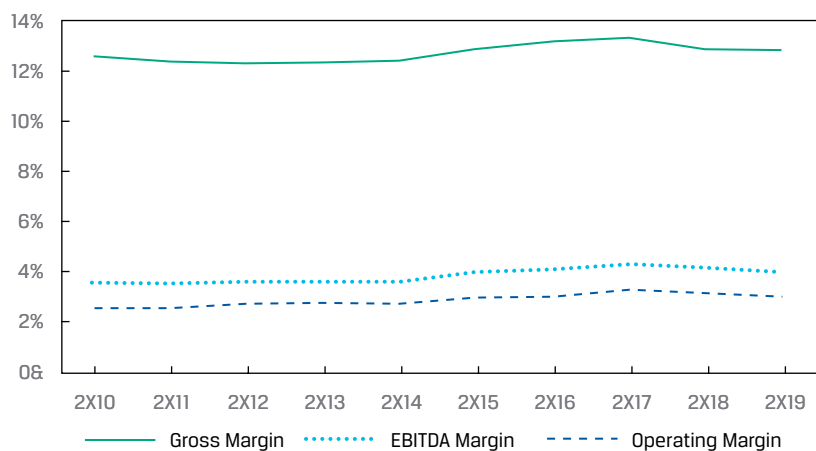
Warehouse Club Inc. Implied Useful Life (line, left) & D&A Expense (bars, right, USD millions)



Warehouse Club Inc. Store Opening Expense per Opening (line, left) & Store Opening Expense (bars, right, USD millions)



Warehouse Club Inc. Operating Profitability

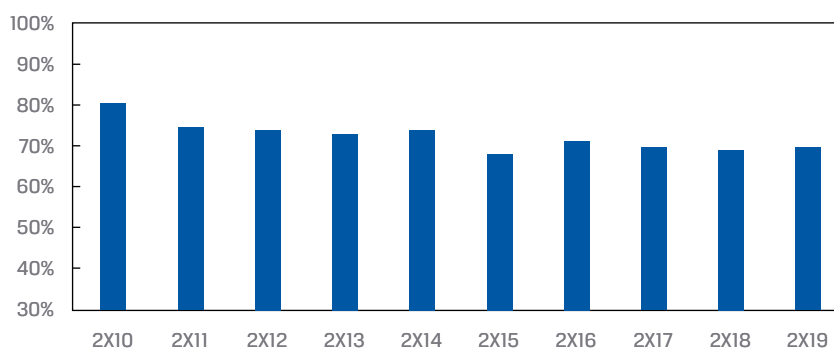


Nguyen makes the following observations:

- Operating costs have increased significantly on an absolute basis from 2X10 to 2X19, but on a relative-to-sales basis have remained stable over the same period, indicating that the driver of the increase has been net sales growth.
- One exception is store opening expenses per store opening, which have quadrupled from 2X10 to 2X19, though in aggregate remain a small cost (less than 1/100 of SG&A expenses). Note that these are operating costs related to store openings, such as employee training before opening, not the investing cost of land and fixed assets.
- SG&A expenses do not appear to be fixed relative to net sales, as they have remained roughly the same percentage of net sales over time.

- Margins have remained mostly stable, though they did increase from 2X15 to 2X17 before coming back down in 2X18 and 2X19. Margins are low for Warehouse, with EBITDA and operating margins of 4% and 3%, respectively, in 2X19.
- The amount of operating profit is similar to the amount of membership fee revenues each year. Nguyen creates the following chart:

Warehouse Club Inc. Membership Fee Revenue as % of Operating Profit



This chart implies that net sales from its stores—after deducting costs of sales, SG&A expenses, D&A, and store opening expenses—account for the remaining 30% of operating profit. This percentage is likely an understatement, since some amount of SG&A expenses is associated with membership fee revenues (marketing, customer service, payment processing, management), but it does align with Warehouse management’s assertion in its annual report that prices provide members with a significant return on membership: the company earns a thin margin on net sales, and it is membership fees that allow the company to run profitably.

Working Capital

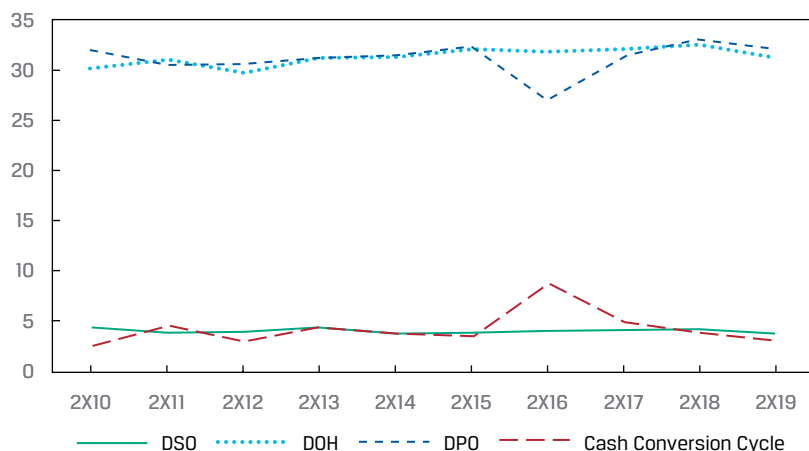
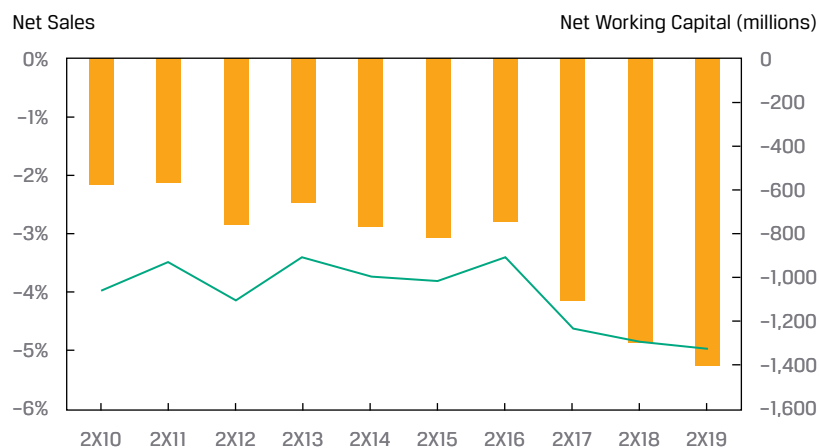
Recall from earlier modules on working capital that the primary measures of a company’s working capital management are activity ratios that determine its cash conversion cycle and the ratio of net working capital to sales. A short cash conversion cycle means that the company requires less external financing to fund operations, while net working capital requirements determine a minimum level of investment, in addition to capital investments, that cannot be distributed to investors. Negative net working capital means that suppliers are a source of financing.

CASE STUDIES

Warehouse Club Inc. Working Capital Management



Elaine Nguyen calculates the working capital measures for Warehouse.

Warehouse Club Inc. Activity Ratios and Cash Conversion Cycle (Days)**Warehouse Club Inc. Net Working Capital to Net Sales (line, left) and Net Working Capital (bars, right, USD millions)**

Nguyen makes the following observations:

- The cash conversion cycle is short, less than a week, primarily as a result of a DSO of 4, because Warehouse's customers pay in cash or with credit or debit cards that settle in less than five business days, and days of inventory on hand is nearly equal to days payable outstanding, indicating that inventory is financed by suppliers.
- Net working capital is negative, now down to -5.0% of net sales, indicating that it is a source of financing for the company.
- Generally, all the activity ratios have been stable since 2010, except for a one-year increase in 2016 from a decline in days payable outstanding. Nguyen looked into this increase in the notes to the financial statements in a historical annual report. The company indicated that it changed accounting information systems in early 2017 and made accelerated payments to suppliers at the end of 2016, in advance of the system change in case there were problems.

QUESTION SET

Questions 1–4 relate to the working capital data below for Iliso Marketplace, as well as to Iliso's financial statements in the lesson "Determining the Business Model".

Working Capital	2X15	2X16	2X17	2X18	2X19
Accounts receivable, net	29	50	50	52	58
<i>DSO</i>	4	5	4	3	3
Inventories, net	190	250	325	384	439
<i>DOH</i>	37	36	32	32	30
Prepayments and other	23	32	35	56	59
<i>% of total revenue</i>	1%	1%	1%	1%	1%
Accounts payable	138	190	272	321	410
<i>DPO</i>	27	28	27	26	28
Taxes payable	2	8	10	12	29
<i>% of total revenue</i>	0%	0%	0%	0%	0%
Accrued expenses and other current	111	193	232	295	354
<i>% of total revenue</i>	4%	6%	4%	5%	5%
Net working capital	(8)	(59)	(104)	(136)	(237)
<i>NWC to sales</i>	0%	–2%	–2%	–2%	–3%
Cash conversion cycle	14	14	9	8	5

1. Compared with 2X18, Iliso's operating margin change in 2X19 is *closest* to:

- A. 0.07%.
- B. 0.57%.
- C. 80.56%.

Solution:

B is correct. Operating (EBIT) margin is Operating profit/Revenue:

$$2X18: 72/6,186 = 1.163\%$$

$$2X19: 130/7,526 = 1.727\%$$

Subtracting 2X18 EBIT margin from 2X19 EBIT margin gives:

$$1.727\% - 1.163\% = 0.564\%$$

A is incorrect, because it is the change in gross profit margin.

C is incorrect, because it is the year-over-year change in operating profit.

2. What is the *most likely* explanation for Iliso's improvement in inventory days on hand from 2X15 to 2X19?

- A. More customers paying in cash
- B. Strong growth in net retail sales

C. Rising proportion of GMV from third-party merchant sales

Solution:

C is correct. With third-party merchant sales, Iliso acts as an agent for the third-party merchant and does not take control of inventory. Accordingly, as the proportion of third-party merchant sales rises in GMV and Iliso's revenues, the portion of revenues associated with inventories falls.

A is incorrect, because there is no indication that Iliso accepts cash on delivery (as is common in China and other geographies). It appears that, like other e-commerce platforms, Iliso accepts only online payments.

B is incorrect, because growth in retail sales is likely to grow accounts receivable as well; days sales outstanding should be unaffected.

3. Which of the following expenses are *most likely* to vary with output?

A. depreciation

B. costs of sales

C. interest expense

Solution:

B is correct. Costs of sales are inventory costs that are recognized when a sale is made.

A is incorrect, because depreciation varies with fixed assets. This expense is a fixed cost in the short run. Recall that fixed operating costs do not change within a given range of output in the short run.

C is incorrect, because interest expense is a fixed cost component that arises from the company's capital structure, not its revenues.

4. Iliso doesn't disclose gross profit on third-party merchant sales. However, an analyst observes that a competing marketplace platform makes 85% gross profit margin on third-party merchant sales. Assuming that Iliso's gross profit margin is similar, what proportion of Iliso's gross profit *most likely* came from third-party merchant sales in 2X19?

A. 7.6%

B. 15.0%

C. 28.1%

Solution:

A is correct.

$$\text{Gross profit of third-party merchant sales} = 85\% \times 190 = \$162$$

$$\begin{aligned} \text{Gross profit of Iliso} &= \text{Total revenues} - \text{Costs of revenues} = \$7,526 - \$5,414 \\ &= \$2,112 \end{aligned}$$

$$\text{Proportion of gross profit arising from merchant sales} = \$162 / \$2,112 = 7.6\%$$

B is incorrect; it is the cost of revenues on third-party merchant sales.

C is incorrect; it is Iliso's gross profit margin in 2X19.

6

CAPITAL INVESTMENTS AND CAPITAL STRUCTURE

- ☐
- evaluate a company’s capital investments and capital structure

Sources and Uses of Capital

Firms invest capital from debt and equity investors to earn returns above their investors’ required rates of return. An important consideration in company analysis is to evaluate whether required rates of return have been met or exceeded (i.e., whether economic value is created for investors) over the longer run and to assess risks and opportunities associated with the capital structure, such as the use of financial leverage. Before evaluation, a first step is to determine a company’s sources and uses of capital on a historical basis, as shown generically in Exhibit 6.

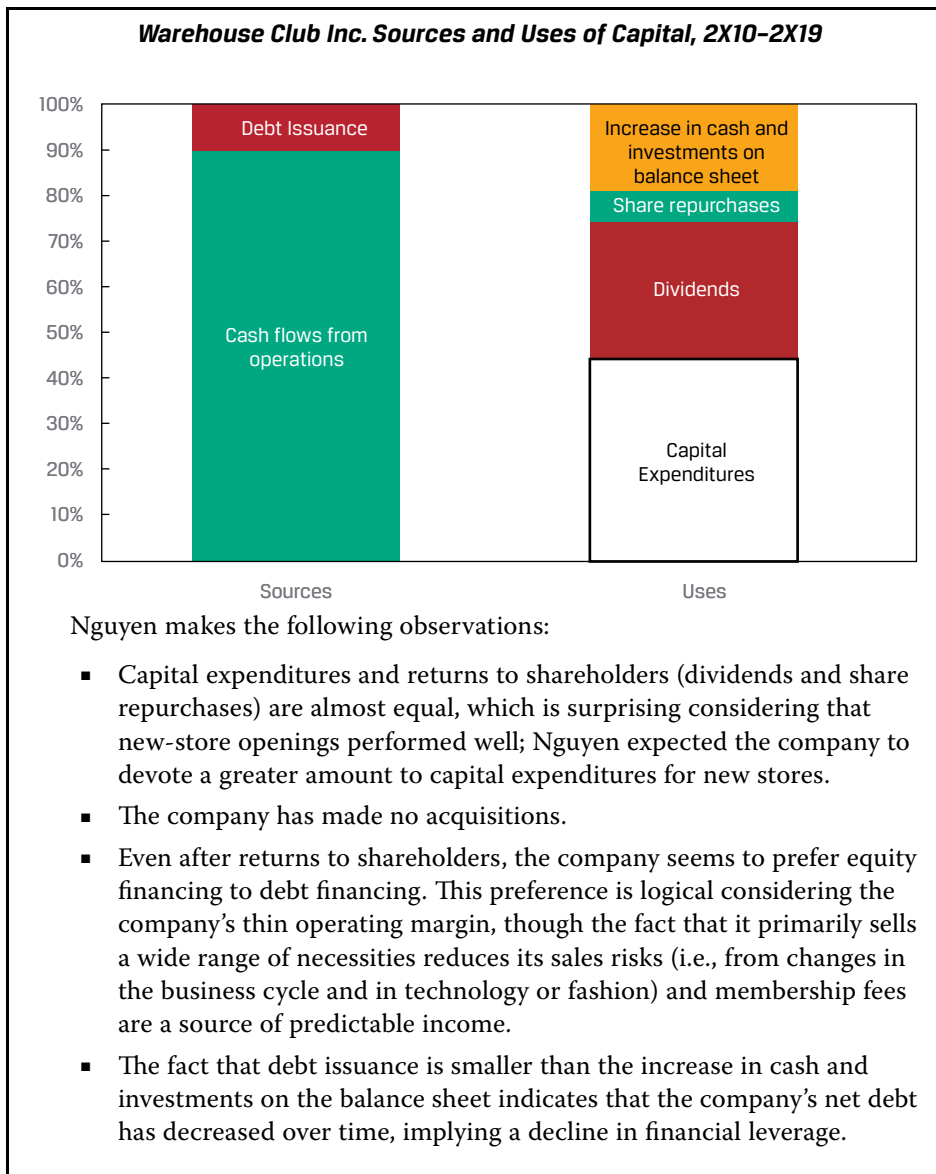
Exhibit 6: Sources and Uses of Capital	
Sources of Capital	Uses of Capital
Cash flows from operations, including net working capital (if negative)	Cash and investments on hand
Debt issuance	Net working capital (if positive)
Equity issuance	Capital expenditures and additions to intangibles
Asset disposals	Acquisitions
	Debt paydown
	Dividends and share repurchases

CASE STUDIES

Warehouse Club Inc.’s Sources and Uses of Capital

Elaine Nguyen calculates the following information on Warehouse over 2X10–2X19. Cash flows from operations include the effect of additional net working capital financing (i.e., Warehouse’s net working capital became more negative over time).





Evaluating Capital Investments and Capital Structure

Returns on invested capital over the longer run, compared with investors' required rates of return, can be used to evaluate whether management has used investors' capital wisely. Recall that independent investment analysts do not have access to the same information that management uses to assess individual projects with IRRs and NPVs, so analysts have to use more aggregated measures to discern whether there have been worsening or improving trends in value creation.

Risks related to the capital structure can be measured using leverage and coverage ratios, credit ratings by third-party rating agencies, and the sensitivity of net income to changes in operating income known as the **degree of financial leverage**, as shown in Equation 3. The degree of financial leverage is increased by higher interest expenses that are fixed with respect to operating income. The degree of financial leverage is analogous to the degree of operating leverage but is determined by financing costs rather than fixed operating costs.

$$DFL = \% \Delta \text{ Net income} / \% \Delta \text{ Operating income} \quad (3)$$

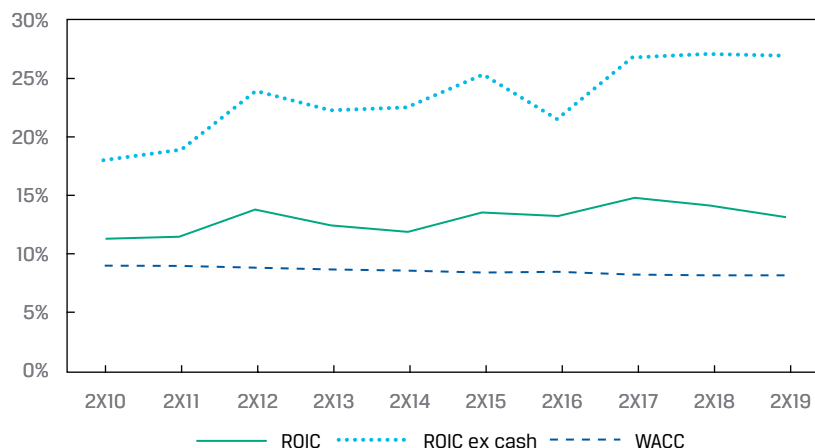
CASE STUDIES

Evaluating Warehouse Club Inc.'s Capital Investments and Capital Structure

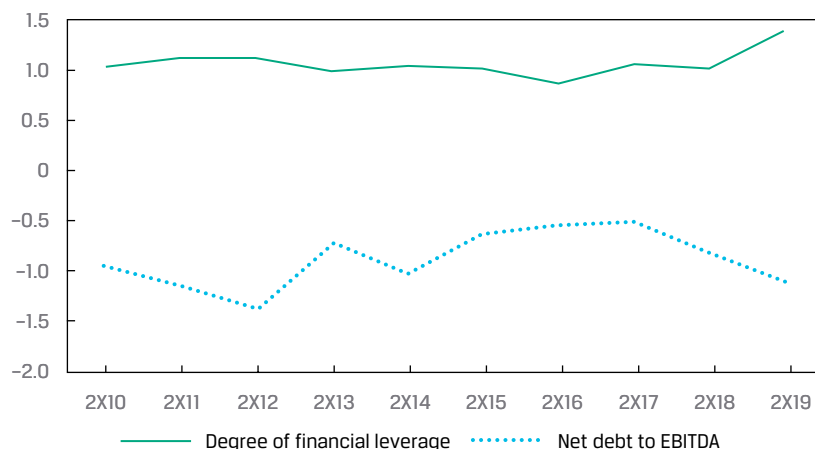


Elaine Nguyen calculates the following information on Warehouse over 2X10–2X19. Additionally, the company's interest coverage ratio (EBIT to interest expense) has been over 20 times and recently reached 30 times in 2X19, and the company's credit is rated investment grade by major credit rating agencies.

Warehouse Club Inc. Return on Invested Capital and Weighted Average Cost of Capital



Warehouse Club Inc. Financial Leverage Ratios

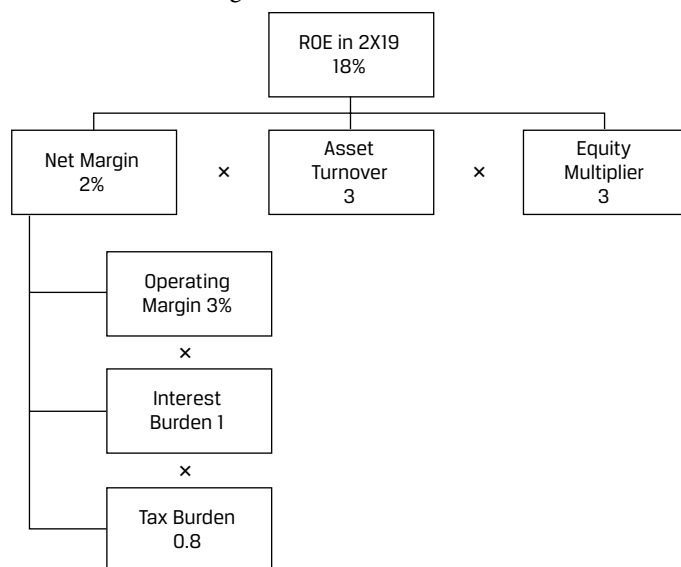


Nguyen makes the following observations:

- The company has created value for investors, with a spread of ROIC over WACC of 200–700 basis points since 2X10. WACC has declined slightly over time as interest rates have fallen; this calculation assumes a constant required rate of return on equity of 9%.
- The company's returns on capital are far higher than its low operating and net margins would suggest, owing to its high asset turnover and negative net working capital as a low-cost means of financing from suppliers, requiring less capital from investors.

- The company has a conservative capital structure, with negative net debt and a degree of financial leverage close to 1.0 because interest expense is largely covered by interest income on cash holdings (i.e., net and operating income move together).
- The capital structure conservatism is evidenced in the counterfactual ROIC, which excludes cash and investments on the balance sheet. While an exaggeration (the company would need *some* cash on hand), ROIC could be almost 15 percentage points higher if the company reduced net cash by investing in value-creating projects or by returning capital.

Unlevered returns, as measured by ROIC or return on assets, are augmented by financial leverage to produce levered returns, or return on equity, which was introduced in an earlier module. We can use return on equity and its decomposition as a comprehensive measure of profitability for an issuer. The ROE decomposition for Warehouse Club Inc. illustrates the drivers of the company's levered returns, which are well in excess of its net margin.



QUESTION SET



Questions 1–4 relate to Iliso's financial statements in the lesson "Determining the Business Model".

1. In 2X19, the market value of Iliso's equity was \$1,178 million, and the book value of debt, which was approximately equal to the market value, was \$144 million. Assuming a cost of debt of 5.5%, cost of equity of 12%, tax rate of 24%, and price-to-book ratio of 1.0, Iliso's WACC is *closest* to:

- A. 11.15%.
- B. 11.30%.
- C. 12.00%.

Solution:

A is correct. Recall that:

$WACC = [\text{weighting of debt} \times \text{gross cost of debt} \times (1 - \text{tax rate})] + (\text{weighting of equity} \times \text{cost of equity})$.

Total capital is \$1,178 + \$144 = \$1,322 million. Weighting of debt is therefore \$144/\$1,322 = 10.89%, and Weighting of equity is 1 – Weight of debt = 89.11%.

Therefore, $WACC = [10.89\% \times 5.5\% \times (1 - 24\%)] + (89.11\% \times 12\%) = 11.15\%$.
B is incorrect, because it incorrectly calculates WACC using a pre-tax rather than after-tax cost of debt.

C is incorrect; it is the cost of equity.

2. Why might Iliso's WACC be significantly higher than Warehouse Club's WACC, as calculated earlier in the module?

- A. Warehouse Club's business model is less risky than e-commerce.
- B. Warehouse Club's capital structure, with degree of financial leverage close to 1.0, is less conservative than Iliso's.
- C. Iliso is an established retailer with an investment-grade rating.

Solution:

A is correct. The cost of equity is the compensation that equity investors demand for the business and financial risks of investing in an issuer's equity securities. Iliso's cost of equity is 12%, whereas Warehouse Club's is 9%, reflecting Warehouse Club's longer operating history, more proven business model, and competitive positioning.

B is incorrect, because Warehouse Club's capital structure, with degree of financial leverage of 1.0, would imply almost 100% weighting of equity, whereas Iliso has an 11% debt weighting and therefore Warehouse's capital structure is *more* conservative.

C is incorrect, because an investment-grade rating would reduce Iliso's cost of debt and WACC.

3. Iliso's degree of financial leverage in 2X19 is *closest* to:

- A. 0.77.
- B. 1.13.
- C. 1.84.

Solution:

B is correct, calculated as Change in net income/Change in operating income = $(100/52 - 1)$ divided by $(130/72 - 1) = 1.146$.

A is incorrect, because it is incorrectly calculated as 2X19 Net income/Operating income = $100/130 = 0.77$.

C is incorrect, because it is Iliso's equity multiplier: Total assets/Equity = $2,168/(592 + 586) = 1.84$.

4. Explain why Iliso's ROE differs from Warehouse Club's ROE.

Solution:

Iliso's ROE is lower than Warehouse Club's ROE.

$ROE = \text{Net margin} \times \text{Asset turnover} \times \text{Equity multiplier}$

$$\text{ROE} = 1.3\% \times 3.5 \times 1.84$$

= 8.37%, which is less than half of Warehouse Club's 18% ROE.

This primarily results from Iliso's lower net margin (1% versus 2%) and an equity multiplier that is 40% lower (1.84 versus 3), which is somewhat offset by Iliso's higher asset turnover (3.5 versus 3).

PRACTICE PROBLEMS

The following information relates to questions 1-7

Joshua Hu, a research analyst, is initiating coverage on several companies in the ocean freight shipping industry. OldShips is a mature company with high fixed costs and a high capital expenditure to sales ratio because it owns and operates its own fleet of ships. CleanYards is a technologically advanced, sustainable shipyard with a focus on specialized repairs and ship construction. Hu has compiled the following data for the two companies:

	OldShips	CleanYards
Degree of Financial Leverage (DFL)	2.0	2.0
Degree of Operating Leverage (DOL)	1.0	2.0
Asset turnover ratio	0.84×	0.42×
DSO	27	98
DOH	12	46
DPO	55	40

NewShips, a third company, is a web-based shipping technology platform that connects ship operators such as OldShips with customers in a wide variety of industries who need ocean freight shipping. NewShips' customers place orders online and pay for freight to be placed on a container to any destination in the world. NewShips' partners, like OldShips, provide vessels on both long- and short-term charters.

In 2X19, NewShips' platform brokered orders for 900,000 twenty-foot equivalent unit (TEU) containers in aggregate, with an average gross freight rate of USD3,848 per TEU. On average, NewShips' commission, which it receives as a broker from the customer, was 6% of the freight rate.

- Which of the following is *most likely* to be an element of Hu's subsequent reports?
 - Industry size, growth rate, and key drivers
 - Summary of changes to the recommendation
 - Ownership structure and management composition
- Shipping is a highly cyclical industry. Which of the following companies *most likely* has the most ability to ramp up capacity to meet increased demand for shipping services?
 - OldShips
 - NewShips
 - CleanYards

3. In 2X19, NewShips' revenue was *closest* to:
 - A. \$173 million.
 - B. \$1.46 billion.
 - C. \$3.63 billion.
 4. Assuming both companies have similar equity multipliers, OldShips is *most likely* to have a:
 - A. much lower ROE than CleanYards.
 - B. similar ROE to CleanYards'.
 - C. much higher ROE than CleanYards.
 5. When conducting company analysis, which of the following is *least* relevant to determining a company's business model?
 - A. Company annual report describing the firm's product lines
 - B. Comments by management about competing products and substitutes
 - C. Industry white papers regarding the company's product pricing strategy
 6. OldShips' cash conversion cycle (CCC) is:
 - A. lower than CleanYards'.
 - B. similar to CleanYards'.
 - C. higher than CleanYards'.
 7. The first step in reviewing a company's capital investments is to:
 - A. evaluate management's investments.
 - B. identify operating cash flow drivers.
 - C. determine the major sources and uses of cash.
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SOLUTIONS

1. B is correct. A subsequent company research report focuses on updating investors on new information or changes in an analyst's price target and recommendation.
A and C are incorrect, because in-depth information regarding industry size, growth rate, and key drivers or ownership structure and management composition is more suited to an initiation report.
2. B is correct. Only NewShips has a capital-light business model. Both OldShips and CleanYards require substantial capital investment in order to increase supply capacity. New ship construction (for OldShips) and acquiring new yards (for CleanYards) take time and cannot be met immediately.
3. A is correct. NewShips acts as a broker and earns 5% of the freight rate in commissions. In 2X19, NewShips brokered 900,000 TEUs at an average freight rate of 3,848 per TEU, earning a 5% commission: $900,000 \times 3,848 \times 0.05 = \173 million.
4. B is correct; $ROE = DTL \times \text{Asset turnover} \times \text{Equity multiplier}$.
OldShips has half the degree of CleanYards' total leverage. However, its asset turnover is double CleanYards' ($0.84\times$ versus $0.42\times$). Since the equity multiplier is the same, their ROEs must be similar.
 $\text{Degree of total leverage (DTL)} = DFL \times DOL = (\% \Delta \text{ Net income} / \% \Delta \text{ Operating income}) \times (\% \Delta \text{ Operating income} / \% \Delta Q)$.
Thus, OldShips' $DTL = 2 \times 1 = 2$, whereas CleanYards' $DTL = 2 \times 2 = 4$.
A and C are incorrect, because both companies must have similar ROEs.
5. B is correct. Competing products and substitutes are analyzed in Porter's Five Forces. This analysis is usually performed to better understand a company's industry and its competitive positioning. Management comments are unlikely to guide the analyst to better understand the company's business model.
A and C are incorrect, because both are relevant sources for an analyst in determining the company's business model.
6. A is correct.
 $CCC = DOH + DSO - DPO$, so:
OldShips' $CCC = 12 + 27 - 55 = -16$ days
CleanYards' $CCC = 46 + 98 - 40 = 104$ days
OldShips, in fact, has a negative cash conversion cycle. OldShips takes advantage of the favorable credit terms granted by its suppliers.
B and C are incorrect, because OldShips' CCC is lower than CleanYards'.
7. C is correct. Before a review of a company's capital investments, the first step is to determine the company's sources and uses of capital on a historical basis.
A and B are incorrect, because both take place after the analyst has identified the company's sources and uses of cash and determined what to focus on.